

Content Disarm and Reconstruction as an Adversarial Defense for Vision-Language Models: An Empirical Evaluation Against Pixel-Space Attacks

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In plain English

We altered 200 photos by amounts too small to see (up to 16 gray levels out of 255) so that a vision-language model, LLaVA-1.5-7B, would answer a question about the photo with a word we chose in advance. With no defense, this worked on 11% of the photos at the weakest setting, 26% at the middle one and 55.5% at the strongest.

Then we ran each altered photo through a defense before the model saw it. The defenses were JPEG re-saving at three qualities and Dangerzone, an open-source tool that sanitizes a file by rendering it to pixels and rebuilding it. Every defense removed almost all of the attack. Of the 185 attacks that worked without a defense, none survived Dangerzone or JPEG quality 50, one survived JPEG quality 75 and two survived JPEG quality 90.

Dangerzone matched JPEG. It stopped the same attacks and changed the model’s answer on clean photos about as often (8.5% against 7.5% to 8%). We found no evidence that the sanitizer protects better than a plain JPEG re-save.

Two cautions apply. The attacker did not know a defense existed, and any lossy re-save seems to break this kind of attack, so the result says little about an attacker who adapts. Also, we measured one sanitizer on one model with one attack. The second sanitizer in the plan, ICDR, could not be run, and typographic attacks were not tested.

1 Introduction and research questions

Vision-language models (VLMs) such as LLaVA [1] pass an image through a CLIP-style vision encoder [2] before the language model reads it. A small, bounded change to the pixels can steer the generated text [3], and re-encoding the input is a common, model-agnostic defense [4]. Separately, the security industry sanitizes files with Content Disarm and Reconstruction (CDR): parse a file, discard everything that is not validated content, and rebuild a clean file. Security guidance already recommends multimedia CDR against multimodal prompt injection [5], and the proposal treats that recommendation as unvalidated for adversarial pixels. This report tests it for one deployed tool.

The research questions come from the proposal:

- **RQ1.** Does a deployed CDR tool reduce the success of a targeted pixel-space attack on a VLM?
- **RQ2.** Does CDR do better than JPEG re-encoding at the same cost to legitimate content?
- **RQ3.** Is any robustness consistent with removal of high-frequency perturbation energy?
- **RQ4.** Can a defense-aware attacker recover the attack?

RQ4 is an extension that the minimum study did not cover (Section 4).

CDR is an existing technique and Dangerzone [6] is an existing tool. The contribution here is the measurement, with paired statistics, on a fixed set of images.

2 Method

2.1 Model, data and attack

The model is LLaVA-1.5-7B (llava-hf/llava-1.5-7b-hf) with its CLIP ViT-L/14-336 vision tower, loaded in 4-bit NF4 with fp16 compute so that it fits a 16 GB GPU, and decoded greedily. The question is “What is the main object in this image? Answer with a single lowercase word.”

The 200 images are 10 per class from 20 ImageNet classes, taken from the Imagenette and Imagewoof subsets and resized to 336×336 . This is a fixed 20-class set, not a random 20 of 1000 classes. Each image has a target label, the next class in sorted order, which differs from its true class.

The attack is targeted L_∞ projected gradient descent [7]: teacher-forced cross-entropy toward the target word, 200 steps of $1/255$, random start, seed 1234, with $\epsilon \in \{4, 8, 16\}/255$. The attacker is defense-unaware. Each adversarial image is crafted once against the undefended model (condition D0), stored losslessly, and then passed through every defense. An attack counts as a success when the model’s actual greedy answer contains the target word. That makes 600 attacks (200×3).

2.2 Defenses

- **D0:** no defense (an 8-bit round trip only).
- **JPEG 90, 75, 50:** Pillow, 4:2:0 subsampling, deterministic.
- **Dangerzone 0.11.0:** the real tool, run through rootless Podman. Dangerzone works on documents, so each image is wrapped in a one-page PDF, sanitized, and rasterized back to 336×336 . The PDF round trip is part of what we measure.

ICDR [8, 9] was planned as a second CDR tool. It is built on a commercial imaging library (Aspose), its committed code does not compile, and no open substitute exists, so it was not run and nothing stands in for it.

Dangerzone could not run on the Colab GPU host, which needs a rootless container runtime. We split the work: the attack and the model answers ran on Colab, and the sanitizing ran on a local machine with rootless Podman. The attacked images were saved as lossless arrays, so the split changes no input. The sanitized images went back to the same 4-bit model for the answers.

2.3 Metrics and statistics

All metrics are relative to the model’s own answers, because a general VLM does not reliably emit fine-grained class names.

- **Targeted ASR:** the share of attacks whose answer contains the target.
- **Preservation:** the share of clean images where the defense leaves the model’s clean answer unchanged. This is the cost of the defense on legitimate input.
- **Restoration:** the share of attacked images where the defended answer equals the model’s clean answer.
- **Mechanism:** for each attacked image, the residual $\text{defended_adv} - \text{defended_clean}$, which removes the defense’s own footprint. We report its energy relative to the undefended residual and its share of high-frequency energy in a radial Fourier spectrum.

Every defense sees the same images and the same adversarial examples, so outcomes are paired by image. We use McNemar’s test on the discordant pairs, 95% percentile bootstrap intervals (10,000 resamples) for rates and for paired differences, and Holm correction across the family of seven comparisons (each defense against D0, and Dangerzone against each JPEG quality). The family is evaluated at the largest budget, $16/255$. For $4/255$ and $8/255$ we ran the exact binomial McNemar test separately, without correction.

Table 1: Targeted ASR in percent with 95% bootstrap interval, $N = 200$ per cell.

Defense	4/255	8/255	16/255
D0 (none)	11.0 [7.0, 15.5]	26.0 [20.0, 32.5]	55.5 [48.5, 62.5]
JPEG 90	0.0	0.5 [0.0, 1.5]	0.5 [0.0, 1.5]
JPEG 75	0.0	0.0	0.5 [0.0, 1.5]
JPEG 50	0.0	0.0	0.0
Dangerzone	0.0	0.0	0.0

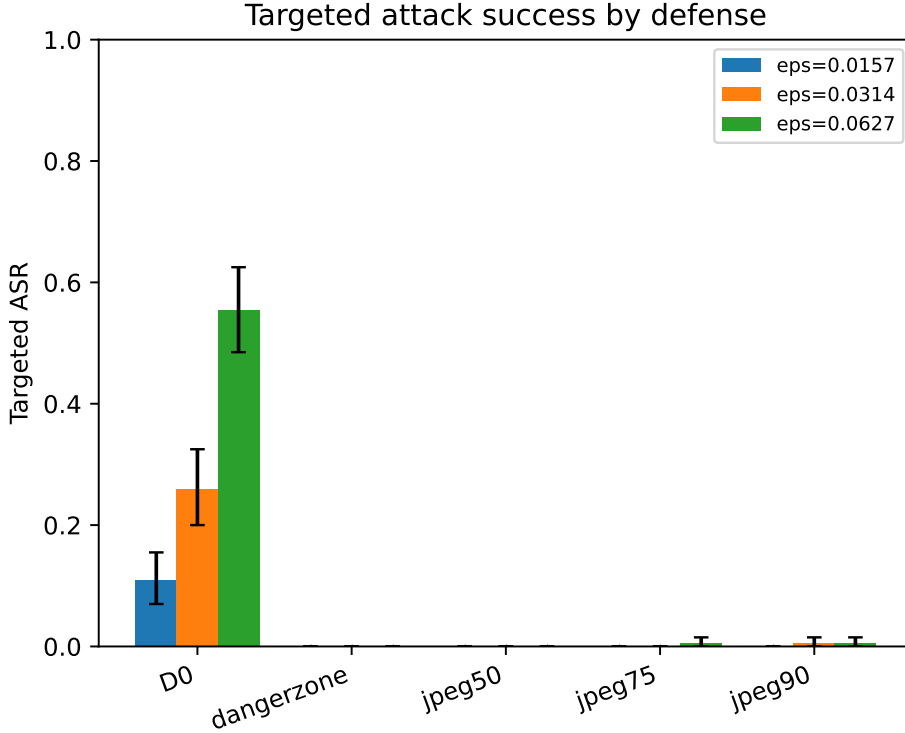


Figure 1: Targeted ASR by defense and budget (error bars: 95% bootstrap interval). The ϵ labels are in $[0, 1]$ units: 0.0157, 0.0314 and 0.0627 are 4, 8 and 16/255.

2.4 Compute

The 600 attacks ran on Colab T4 GPUs at about 285 to 330 seconds each, roughly 50 GPU hours, spread over sessions of at most three hours. State lived on Drive and each session skipped finished jobs. The attack was never recomputed. Sanitizing 800 images (600 attacked, 200 clean) took about 47 minutes locally. The run finished with 3000 defense evaluations and no failures. The experiment code is at commit 5792229.

3 Results

3.1 The attack works, and every defense removes it

Table 1 gives targeted ASR. Without a defense it rises from 11% to 55.5% as the budget grows. Under any defense it is at or near zero at every budget. Where the count is zero, the bootstrap interval collapses to a point, so a zero means none in 200 trials, and the rule of three puts the 95% upper bound near 1.5%.

Of the 185 attacks that succeeded without a defense (22, 52 and 111 at the three budgets), Dangerzone and JPEG 50 neutralized all of them, JPEG 75 left one and JPEG 90 left two.

Table 2: Paired comparisons at 16/255. (b, c) counts the images where only the first or only the second condition was attacked successfully. The ASR difference is first minus second, in percentage points, with a 95% paired bootstrap interval.

Comparison	(b, c)	ASR difference	Holm-adjusted p
Dangerzone vs D0	(0, 111)	-55.5 [-62.5, -48.5]	1.1×10^{-24}
JPEG 50 vs D0	(0, 111)	-55.5 [-62.5, -48.5]	1.1×10^{-24}
JPEG 75 vs D0	(0, 110)	-55.0 [-62.0, -48.0]	1.3×10^{-24}
JPEG 90 vs D0	(0, 110)	-55.0 [-62.0, -48.0]	1.3×10^{-24}
Dangerzone vs JPEG 50	(0, 0)	0.0 [0.0, 0.0]	1.0
Dangerzone vs JPEG 75	(0, 1)	-0.5 [-1.5, 0.0]	1.0
Dangerzone vs JPEG 90	(0, 1)	-0.5 [-1.5, 0.0]	1.0

Table 3: Preservation (clean images, percent, 95% interval) and restoration (attacked images, percent) per budget.

Defense	Preservation		Restoration		
			4/255	8/255	16/255
D0 (none)		100.0	49.5	35.5	17.0
JPEG 90	92.5	[88.5, 96.0]	84.5	77.0	68.5
JPEG 75	92.0	[88.0, 95.5]	88.5	85.0	78.5
JPEG 50	92.0	[88.0, 95.5]	89.5	88.5	83.0
Dangerzone	91.5	[87.5, 95.0]	88.0	87.5	81.5

No defense produced a success on an image where D0 failed. Table 2 gives the paired tests at 16/255, where the family of seven comparisons is corrected with Holm. All four defenses differ from D0 with adjusted $p < 10^{-23}$. At 4/255 and 8/255 every defense also beats D0 on the exact test (uncorrected $p \leq 4.8 \times 10^{-7}$ at 4/255 and $p \leq 8.9 \times 10^{-16}$ at 8/255, which passes a Bonferroni threshold for twelve tests).

3.2 Cost to legitimate content

Table 3 gives preservation and restoration. Preservation is about 92% for every defense, and the intervals overlap fully: Dangerzone is at 91.5% [87.5, 95.0], JPEG 50 and 75 at 92.0% and JPEG 90 at 92.5%. Quality 90 costs as much as quality 50, which suggests that much of the 7.5% to 8.5% of changed clean answers comes from the greedy 4-bit model flipping under small input changes. We did not test that cause directly.

Restoration shows a difference between the JPEG qualities. At 16/255 it is 81.5% [76.0, 87.0] for Dangerzone, 83.0% [77.5, 88.0] for JPEG 50, 78.5% [72.5, 84.0] for JPEG 75 and 68.5% [62.0, 75.0] for JPEG 90, against 17.0% [12.0, 22.5] with no defense. A weak JPEG usually moves the answer away from the target without returning to the clean answer. Dangerzone and JPEG 50 return to the clean answer more often, and their intervals overlap.

3.3 What the defenses do to the perturbation

Table 4 and Figure 3 show the defense-controlled residual. Dangerzone removes the most: the residual keeps 59% of its energy and the high-frequency share falls from 0.82 to 0.61. JPEG 50 lowers the high-frequency share to 0.73 but raises the residual energy by 26%, and JPEG 75 does the same to a smaller degree. JPEG 90 leaves the energy at 0.99 of the original and the high-frequency share at 0.85, slightly above where it started.



Figure 2: Security and utility at 16/255. The vertical axis is $1 - \text{ASR}$ and the horizontal axis is preservation. The three JPEG points and Dangerzone sit in the same corner.

Table 4: Mean residual energy ratio and high-frequency share of the residual defended_adv – defended_clean, over 588 attacked images per defense. The other 12 images were processed by the pilot with an earlier metric and are excluded.

Defense	Energy ratio (after / before)	High-freq. share before	after
D0 (none)	1.00	0.82	0.82
JPEG 90	0.99	0.82	0.85
JPEG 75	1.14	0.82	0.80
JPEG 50	1.26	0.82	0.73
Dangerzone	0.59	0.82	0.61

4 Discussion

RQ1: yes. Against this attacker, every defense removes nearly all of the attack (Table 1). At 16/255 the ASR falls from 55.5% to between 0% and 0.5%, and the paired tests are significant after correction.

RQ2: no evidence that CDR beats JPEG. Dangerzone and JPEG 50 stopped the same attacks (zero discordant pairs), and Dangerzone against JPEG 75 and 90 differs by at most one image of 200 (adjusted $p = 1.0$). Preservation is the same within the intervals (Figure 2), and restoration overlaps with JPEG 50. The hypothesis that Dangerzone lies beyond the JPEG trade-off curve is not supported. A small advantage cannot be excluded, but with ASR at the floor the test has no room to show one. This is a limit of the attack, taken up below.

RQ3: removal of high-frequency energy is not needed here. Dangerzone removes the most high-frequency energy, as expected for a render-and-rebuild pipeline. Yet JPEG 90 defeats 183 of 185 attacks while leaving the residual energy unchanged and the high-frequency share slightly higher. A defense can break this attack without removing the perturbation’s energy. One reading is that the perturbation is tuned to the exact pixels the model saw, and any small resampling shift breaks the optimization. We did not test that reading, and the mechanism numbers are descriptive.

RQ4: not tested. The attacker never saw a defense. The finding that JPEG 90, which

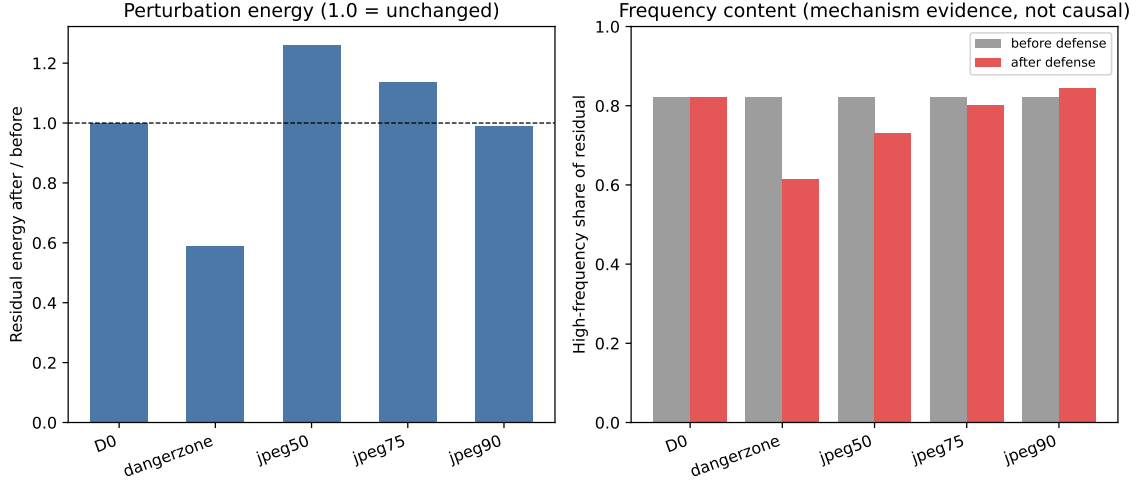


Figure 3: Residual energy ratio (left) and high-frequency share (right) per defense.

changes the image least, already defeats the attack suggests that the attack is brittle. The informative next experiment is a defense-aware attack that approximates the defense in the backward pass or averages over its randomness [10–12]. Without it, these results do not support the claim that CDR protects a VLM.

5 Limitations

- **One CDR tool.** Only Dangerzone, through an image-to-PDF adapter, was measured. ICDR was blocked. The results do not speak for CDR as a category.
- **Non-adaptive attacker and a floor effect.** With ASR near zero under every defense, the study cannot rank the defenses against each other.
- **Moderate attack strength.** Undefended ASR is 11% to 55.5%. A defense that removes this attack has not been shown to remove a stronger one.
- **One model, one prompt, one data set.** LLaVA-1.5-7B in 4-bit, one question, 200 images from 20 ImageNet classes at 336 px. Quantization changes both the gradients and the answers.
- **Model-relative metrics.** Preservation and restoration compare with the model’s own clean answer, not with ground truth.
- **Scope.** The proposal also names typographic prompt-injection attacks. They were not run.
- **Statistics.** The corrected family is evaluated at 16/255 only. The tests at the other budgets are uncorrected exact tests computed outside the main pipeline. Twelve images in the Dangerzone rows come from the earlier pilot, with the same images, budget and procedure.
- **Harm.** Success means forcing a harmless target word. No harmful content was generated.

6 Reproducibility

The run directory `results/runs/full_200` holds the configuration, environment (Colab T4, torch 2.11, transformers 5.18), dataset manifest, per-job outputs, the persisted adversarial arrays, `statistics.json` and the figures. The seed is 1234 throughout. The scripts, all under `scripts/`:

- `run_experiment.py` runs the attacks and the JPEG conditions and resumes by skipping finished jobs.
- `dangerzone_sanitiz.py` and `dangerzone_finalize.py` run the Dangerzone condition.
- `run_analysis.py` and `make_figures.py` produce every table and figure from the stored results.

Every number in this report is read from those files.

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